

Anchor, Tensor, and Skin: A Fixed-Domain Decomposition Theorem for Admissible Description

Canonical reconstruction, role refinement, and operator identity

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Abstract

Admissible description must distinguish the conditions fixing its original problem, the content that determines its required answers, and the realizations that carry that content. We establish this distinction as a canonical decomposition. For an independently interpreted descriptive carrier, its context map and complete original-answer profile determine a tower of surjections. The carrier is canonically and without representative selection the dependent sum of its realization fibres over that tower. Every minimal complete presentation yields the same middle space by a unique commuting bijection; every faithful reconstruction over a minimal complete content space yields the same dependent fibre structure. Specified variations have exactly three effects: anchor change, tensor change within fixed anchors, or skin change within fixed content. Arbitrary components admit a unique coarsest refinement into homogeneous role incidences. Exact deletion and reconstruction criteria distinguish functional minimality from a claim of three independent Cartesian factors. We further characterize all content-preserving realization maps, prove their compatibility conditions for joint construction, and establish an inspectable-envelope theorem for operator identity under finite sequential use. Worked examples and countermodels show what is lost when a context, required answer, joint domain, or realization distinction is omitted. Exhaustion and impossibility are established through explicit failure witnesses, with original-use coverage, component homogeneity, reconstruction, and the conditions for independence established at their respective levels.

1 The decomposition problem

An engineering record, a mathematical expression, and an operator specification each carry more than a verdict. Their terms must have fixed meanings; their content must answer the questions for which they are used; and some determinate description must present that content. Anchor, tensor, and skin name these three kinds of work. This paper constructs their canonical decomposition and establishes its uniqueness.

1.1 One familiar construction

Consider a record of a bridge trial. The original question specifies the bridge, loading protocol, units, and trial occurrence. Those fixed meanings and conditions do anchor work. The recorded

load, outcome, and any required answer about repeat trials or permitted reuse contribute tensor content. A diagram, a table, and two independently derived calculations can present the same content. Their differences are skin for those original uses when every required answer agrees.

The distinctions can be tested. Replacing a kilogram by a pound while retaining the same numeral changes the interpreted problem unless a lawful unit translation is supplied. Changing the measured load or the answer about a repeat trial changes content in the same problem. Reformatting the record while preserving its meanings, answers, and availability changes skin. A finger supporting the bridge changes a condition of the physical trial; calling that support a feature of the diagram does not conserve the original trial result. A repaired bridge can pass a new trial, while the earlier failed occurrence remains failed.

The record also reveals a difficulty for a theorem about arbitrary “elements.” One clause can specify both a unit convention and a measured value. Replacing that clause in different ways can change the problem, change its content, or merely re-express both. No classification of an individual replacement proves that this entire clause has one exclusive role. The appropriate decomposition must identify the evaluated contribution and its comparison conditions.

1.2 The theorem and its strength

The central construction is

$$D \xrightarrow{P} \coprod_{k \in K} Q_k \xrightarrow{\pi} K, \quad D \cong \coprod_{k \in K} \coprod_{q \in Q_k} R_{k,q}.$$

Here D is the original descriptive carrier, k records its interpreted anchor context, q is the complete profile of original answers within that context, and $R_{k,q}$ is the family of realizations of that content. The symbol $\coprod$ denotes a disjoint union; in the second display the available realization family depends on both preceding entries. Here “tensor” names the complete original-answer content within fixed anchors. No vector-space structure, multilinearity, or tensor-product operation is presupposed by that name. The inverse reconstruction maps are explicit. They do not require choosing a preferred representative in any fibre.

Three different uniqueness claims are proved at their proper levels. The minimal complete content space is unique by a commuting bijection. A faithful realization decomposition is unique as a dependent family over that space. The minimal classification of variation effects is unique on its realized role classes. These results do not identify different occurrences, prescribe an arbitrary partition of written fields, or make incompatible component combinations possible.

Every correctly interpreted description has this canonical organization; every original-use-preserving consumer descends through its content; every specified variation falls into the three effect cases; and every coarse component admits the canonical homogeneous refinement proved below. A proposed alternative either preserves this organization, loses an original distinction, retains extra realization information, or changes the original problem. Particular claims of three nontrivial roles, exclusive raw components, or independent coordinates are then decided by exact tests.

1.3 Proof map and four tests

The argument proceeds through seven dependent steps:

1. Fix the original interpreted context, questions, and bearer before introducing a classifier (Section 2).
2. Construct the canonical context–content–realization tower and its universal property (Section 3).
3. Classify variation effects and construct the coarsest homogeneous refinement of component incidences (Section 4).
4. Prove deletion, minimality, reconstruction, and independent-coordinate criteria (Section 5).
5. Characterize skin maps and prove the extra joint-domain conditions for composition (Section 6).
6. Establish operator identity through an inspectable semantic envelope and its finite uses (Section 7).
7. Apply the tests, localize failures by witnesses, and state the exact downstream exports (Sections 8 to 10).

Four tests recur. The *fibre test* asks whether a retained code merges descriptions with different required answers. The *homogeneity test* asks whether every declared nontrivial variation of a component has the same role. The *reconstruction test* checks the two inverse identities on the stated compatible domain. The *envelope test* asks whether the proposed operator record preserves all original uses, including applicability and any required continuation. A failure has an explicit witness; an assertion of completeness has a corresponding proof obligation.

1.4 Place in the foundation chain

Non-degenerate determinate objects necessarily incur Admissibility, Standing, Reference, and Irreversibility. This paper does not make that governance conditional on adopting a representation. The Kernel identifies this necessary work [1]. The AMetric paper establishes the pre-fixational, symmetry, and non-transmission constraints [2]. The Structure paper constructs complete interfaces and the greatest admissible interior for an independently specified reuse contract [3]. Exhaustion proves original-task coverage, composition of generated uses, and failure alternatives [4]. The present paper establishes the role decomposition and its precise identity, minimality, and component consequences.

The dependency order is one-way: the general Exhaustion proof is independent of the present role theorem. Its quotient and composition results supply the basis for the ATS reconstruction and classification results. Necessary kernel governance applies to the original descriptions, records, and endpoints. The mathematical hypotheses below concern faithful representations of their particular uses.

1.5 Relation to existing mathematics

The constructions use familiar mathematics deliberately. Dependent pairs provide a precise way to retain a realization together with the context and content on which its type depends [5]. Behavioural equivalence and the observable behaviour of state-based systems have a broad treatment in universal coalgebra [6]. Quotient lenses distinguish equality of concrete presentations from equivalence of their interpreted content, and require laws for the associated transformations [7]. These neighbours help state what reconstruction and presentation preservation require; they do not independently establish the AASC necessity claim.

The distinctive result here combines canonical reconstruction, the refinement of mixed component effects, exact minimality witnesses, joint compatibility, and operator-envelope coverage in one dependency theorem. The quotient universal property and the expression of a set as the disjoint union of its fibres supply the standard mathematical basis for these results. The boundary between finite safety and eventual-success questions is retained throughout [8].

2 Original meanings and the anchor context

2.1 The subject precedes the proposed representation

An original task fixes which bearer is described, what counts as the same question, what an answer means, and which uses are claimed. These are not specified by whichever compressed record a proposal happens to retain. In mathematics they may be given by a type, an interpretation, and a family of properties; in an experimental record by the object, occurrence, protocol, and interpretation of its measurements. A proposed description can fail an original requirement while remaining a determinate object whose failure is itself assessable.

We work in classical set theory. All indexed families are sets. No finiteness, decidability, selected representative, metric, or global ordering of admissibility levels is assumed. “Complete” below always names a particular family of original uses. If a smaller language is offered for that family, its coverage must be proved. This stipulation does not supply the family by naming it: the semantic coordinates must already have independently fixed interpretations.

Definition 2.1 (Original descriptive problem). Let D be a set of determinate interpreted descriptions and let $\kappa : D \rightarrow K$ record their original contexts. Replace K by $\kappa(D)$, so every context is attained. Put $D_k = \kappa^{-1}(k)$. For each k , fix a set J_k of original questions and partial original answer functions

$$a_{k,j}^\circ : D_k \rightharpoonup O_{k,j} \quad (j \in J_k).$$

Their totalizations $a_{k,j} : D_k \rightarrow O_{k,j}^\perp$ record either the defined answer or a separate undefined marker. Define

$$p_k(d) = (a_{k,j}(d))_{j \in J_k}, \quad Q_k = p_k(D_k), \quad P(d) = (\kappa(d), p_{\kappa(d)}(d)).$$

An undefined marker is distinct from a negative answer. If different forms of failure or absence are originally distinguished, they receive distinct coordinates or distinct answer values.

The profile p_k is a mathematical object even when no finite procedure computes it. A bivalent qualification is one possible coordinate. A family of bivalent coordinates can have many profiles; two descriptions can both qualify and differ in a required continuation. Nothing in bivalence reduces the complete profile to one bit.

2.2 What an anchor records

An anchor context contains the conditions that fix the interpreted problem: the original types, reference conventions, identities of questions, and governing contract. The context may be represented in many ways. Its identity here is semantic; a lawful recoding with a proved interpretation correspondence is already identified or is explicitly transported before comparison.

The fixed-domain claim takes place within this independently fixed comparison arrangement. Holding k fixed means that an inner construction does not rewrite the governing contract. Comparisons

between different k diagnose a change of problem; they do not license such a change as an operation within either problem. There may be only one k . In that case anchor work remains presupposed, although no anchor-changing variation is realized within D .

A failed application under an unchanged contract must be distinguished from changing the contract. If the fixed question asks whether a proposed bridge withstands a stated load, success and failure are answer values within one context. Similarly, a fixed typing question can have an ill-typed candidate as its negatively classified subject. Changing a required answer, or its availability under a fixed question, changes the profile. It does not by itself prove that an anchor moved. This distinction prevents every ordinary failure from being mislabeled a change of original problem.

The necessary kernel conditions and the AMetric non-transmissive boundary are not adjustable parameters of an interior calculation. Comparing two typed problem specifications in a metatheory does not describe a motion of the pre-metric boundary. Conversely, a metric calibration, licensing record, or physical protocol can vary within a suitably specified larger original task. The role of the variation is determined by what that task holds fixed. This is a difference in the object of comparison, not a permission to erase an earlier incidence.

Proposition 2.2 (Faithful context transport). *Suppose two problem presentations have bijections $f : D_k \rightarrow D'_{k'}$, $u : J_k \rightarrow J'_{k'}$, and bijections $v_j : O_{k,j}^\perp \rightarrow O'_{k',u(j)}^\perp$ preserving the undefined marker, with*

$$a'_{k',u(j)}(f(d)) = v_j(a_{k,j}(d)) \quad \text{for every } d, j.$$

Then f preserves and reflects profile equality. It induces a bijection $Q_k \rightarrow Q'_{k'}$ and restricts to a bijection on each corresponding realization fibre.

Proof. Equality of all coordinates on the left follows from equality of all coordinates on the right. Conversely use the injectivity of each v_j and surjectivity of u . Thus $[p_k(d) \mapsto p'_{k'}(f(d))]$ is well-defined and injective; surjectivity follows from f . The restriction of f to a fibre is bijective because both f and the induced profile map are bijective. $\square$

This is an explicit sufficient transport certificate. One can use more general interpretations, but must then prove the corresponding preservation, reflection, and coverage clauses. A change of notation is not evidence of task change merely because its raw symbols differ. A change of meaning is not faithful transport merely because the same symbols are retained.

2.3 Three levels of identity

The equality on D specifies the descriptive objects being reconstructed. It may be raw syntax, a recorded construction, or an already justified identity quotient. This choice is made before decomposition. Profile equality identifies descriptions agreeing on the original uses. Context equality identifies the original problem in which their answers are read. These three relations can coincide on a special carrier; generally they do not.

If the identity of a particular record occurrence is an original question, include it in J_k . Two distinct occurrences then need not be profile-equivalent even if their measured values agree. If an experiment only asks for a numerical readout, equal readouts alone do not establish identical occurrences. The same discipline applies to proof objects, implementations, and operators. Declaring a syntactic difference non-authoritative for an extensional question does not declare the syntax indeterminate or outside the kernel.

2.4 Coverage and original joint uses

All uses whose preservation is asserted belong to the original task. These can include present answers, availability, retained histories, future tests, and compatibility with other descriptions. One way to obtain a unary complete profile for a joint use is to include all its fixed-context tests: for each permitted external tuple z , retain the question about joint use with z . Such a family can be large. An arbitrary small list of unary features need not cover it.

Exhaustion, Section 4, supplies a compositional route: prove descent of the specified primitive domains and outputs, then prove that the original questions have interpretation-preserving expressions in the generated language. The original family can also contain an infinitary coordinate directly. Finite formation is a theorem about a stated syntax, not a demonstration that every original question has finite observational depth. We keep both options available.

3 Canonical decomposition and its uniqueness

3.1 The dependent realization theorem

For the original data in definition 2.1, put

$$T = \coprod_{k \in K} Q_k, \quad R_{k,q} = \{d \in D_k : p_k(d) = q\}, \quad \mathcal{R} = \coprod_{k \in K} \coprod_{q \in Q_k} R_{k,q}.$$

Every displayed fibre is nonempty by the attained-image definition of Q_k . A member of $\mathcal{R}$ is a triple (k, q, r) whose third entry has the required context and content. The membership requirement is essential: the factors are not asserted to vary independently.

Theorem 3.1 (Canonical ATS reconstruction). *The maps*

$$\Delta : D \rightarrow \mathcal{R}, \quad \Delta(d) = (\kappa(d), p_{\kappa(d)}(d), d), \quad A : \mathcal{R} \rightarrow D, \quad A(k, q, r) = r$$

are mutually inverse. Under this bijection, $P : D \rightarrow T$ forgets the realization within its content fibre, and $\kappa : D \rightarrow K$ forgets both content and realization. The tower

$$D \xrightarrow{P} T \xrightarrow{\pi} K, \quad \pi(k, q) = k,$$

is surjective, with $\kappa = \pi P$.

Proof. For $d \in D$, the triple $\Delta(d)$ has its third entry in the claimed fibre by definition, and $A\Delta(d) = d$. If $(k, q, r) \in \mathcal{R}$, then $\kappa(r) = k$ and $p_k(r) = q$, whence $\Delta A(k, q, r) = (k, q, r)$. Thus both inverse identities hold on their declared domains. The maps P and π are surjective because Q_k and K are attained images. The composite equality follows directly. $\square$

The reconstruction retains the original description in its third entry and requires no selection of a representative from each class. A compressed, independently parametrized skin coordinate requires a separate bijection for each fibre.

3.2 The complete and minimal middle space

Write $d \equiv e$ when $P(d) = P(e)$. The middle space T is the quotient of D by this equivalence, presented as attained profiles. A representation can retain more information than this quotient and still answer every original question correctly. Its surplus becomes important when one claims uniqueness.

Definition 3.2 (Complete and minimal content presentation). A content presentation is a surjection $b : D \rightarrow B$ with a context reader $\beta : B \rightarrow K$ satisfying $\beta b = \kappa$. It is *complete* if every original answer factors through b within its context. It is *minimal* if it is complete and

$$P(d) = P(e) \implies b(d) = b(e).$$

Minimality means that b retains no distinctions between descriptions that agree on context and every original answer. It concerns retained information, not the number of printed fields.

Theorem 3.3 (Universal content and exact uniqueness). *For every complete presentation b , there is a unique surjection $u : B \rightarrow T$ such that $ub = P$. It commutes with context readers: $\pi u = \beta$. It is bijective if and only if b is minimal. In particular any two minimal complete presentations have a unique bijection commuting with their maps from D and with context.*

Proof. If $b(d) = b(e)$, the context reader gives $\kappa(d) = \kappa(e) = k$ and each factored original coordinate gives $a_{k,j}(d) = a_{k,j}(e)$. Hence $P(d) = P(e)$. Set $u(b(d)) = P(d)$. This is well-defined, and surjectivity of b makes it the only possible factor. Surjectivity of P makes u surjective. The context identity holds on each $b(d)$, hence everywhere. If b is minimal, $u(b(d)) = u(b(e))$ implies $P(d) = P(e)$ and then $b(d) = b(e)$, so u is injective. Conversely injectivity of u gives minimality from $ub = P$. For two minimal presentations use the composite of their bijections with T ; commutation with the surjections from D proves uniqueness. $\square$

This proof separates complete adequacy from minimality. A correct representation may retain an extra serial number, a derivation, or a cache. Its existence does not create a new original-answer distinction. Erasing the extra information can be legitimate for the stated questions, while reconstructing the original enriched record requires retaining it in the realization fibre.

Corollary 3.4 (Unique dependent realization structure). *Suppose a minimal complete presentation $b : D \rightarrow B$ is given, and a family $(S_z)_{z \in B}$ has a bijective assembly*

$$a : \coprod_{z \in B} S_z \longrightarrow D, \quad b(a(z, s)) = z.$$

For the unique bijection $u : B \rightarrow T$ of theorem 3.3, the restriction $s \mapsto a(z, s)$ is a bijection $S_z \rightarrow R_{u(z)}$. These maps give the unique isomorphism of the two dependent realizations that commutes with assembly into D .

Proof. The context-content identity $Pa(z, s) = ub(a(z, s)) = u(z)$ places the image in $R_{u(z)}$. Injectivity follows from injectivity of a . If $d \in R_{u(z)}$, surjectivity of a gives $d = a(z', s')$. Then $u(z') = P(d) = u(z)$ and injectivity of u gives $z' = z$. Thus the restriction is surjective. Any commuting isomorphism must take (z, s) to the canonical triple determined by $a(z, s)$, which proves uniqueness. $\square$

Uniqueness here is relative to the given assemblies. An abstract fibre can have many automorphisms. The commuting condition prevents an arbitrary relabeling of its inhabitants from being called a second, physically selected decomposition.

3.3 Consumers and absence of a fourth source of authority

Theorem 3.5 (All original-use-preserving consumers). *Let $F : D \rightarrow Y$ and let $F^\perp$ be its totalization. There is a unique function $\bar{F} : T \rightarrow Y^\perp$ with $F^\perp = \bar{F}P$ if and only if $F^\perp$ is constant on every realization fibre. If constancy fails, two descriptions in one fibre have different definedness or different required results. No postprocessing of P can produce $F^\perp$ on both.*

Proof. If a factor exists, equal inputs under P have equal outputs. Conversely define $\bar{F}(P(d)) = F^\perp(d)$; constancy gives well-definedness and surjectivity of P gives uniqueness. Failure of constancy supplies the stated pair. A function of the same retained input must have the same value on that pair and therefore cannot equal both different original values. $\square$

Every original coordinate satisfies the criterion by construction. A claimed additional original use must either have a proved factor, have a fibre-collision witness, or have its coverage still unestablished. An additional label, level, selector, or classification cannot settle that question by its name. A refinement of P can preserve all original answers and record extra presentation information; it becomes a new authority over the same answers only if a required reader uses that information. Such a use either contradicts fibre constancy or shows that the supposedly complete original profile omitted a required question.

This is the exact no-fourth-authority consequence. It allows mathematics, algorithms, and physical constructions to be discovered or performed. It prohibits claiming that a changed original answer followed solely from retaining the same complete original content. It also permits a new problem with additional original questions; that problem has its own context and completeness proof.

3.4 Qualification, identity, and occurrence

If the only original question is a Boolean qualification b , then the middle space has at most two content classes. If further original questions matter, their coordinates remain. A single positive class can contain many realizations, and even a singleton middle space can have a large realization fibre. Thus unique content, a unique classifying role, and one actual occupant are distinct conclusions. Exhaustion, Theorem 8.1 and Corollary 8.2, give the exact readout and identity-separation conditions for the last step. The reconstruction theorem above retains all occupants and supplies no selection among them.

4 Variation effects and canonical component typing

The dependent reconstruction separates a governing context, complete original content, and a realization of that content. A raw component need not vary in only one of those directions. The appropriate unit of exclusive typing is therefore a specified interpreted variation. We first classify those variations, then construct the unique smallest refinement that gives a homogeneous effect to every component class. This provides component typing without assuming that an arbitrary symbol, field, or clause was already a pure component.

4.1 Original variation incidences

Retain D , κ , p_k , and $P(d) = (\kappa(d), p_{\kappa(d)}(d))$ from the reconstruction. Let C be a set of raw component names and let

$$V \subseteq \{(c, x, y) \in C \times D \times D : x \neq y\}$$

be an independently specified family of interpreted variation incidences. An incidence records the component being varied and its actual before and after descriptions. Its formation conditions may restrict the surrounding context and the permitted replacements. Let $\pi : V \rightarrow C$ be the component projection. Lawful transport is applied before comparing the original task and outcome meanings.

This family is part of the variation problem, not a list selected after classification to remove inconvenient examples. The restriction $x \neq y$ excludes an unchanged description from serving as a witness of realized representational variation. A component with no such incidence has no realized effect in this family.

Definition 4.1 (Effect of an interpreted variation). For $v = (c, x, y) \in V$, define $\epsilon(v) \in \{A, T, S\}$ by

$$\epsilon(v) = \begin{cases} A, & \kappa(x) \neq \kappa(y), \\ T, & \kappa(x) = \kappa(y) = k \text{ and } p_k(x) \neq p_k(y), \\ S, & P(x) = P(y). \end{cases}$$

The names are *anchor effect*, *tensor effect*, and *skin effect*, respectively.

The classification records the first level at which the compared descriptions differ in the tower. An anchor-effect variation may also alter numerical answers or presentation details; those answers belong to different contexts and are not thereby a second tensor effect within one fixed context. Exclusivity concerns the three displayed comparison cases, not a claim that only one visible feature changes.

Changing the applicability answer to an unchanged original question need not change its governing context. If both descriptions have an original interpretation, an applicability difference is a coordinate difference in their complete totalized profiles. For example, a well-typed and an untypable term can be assessed by the same original typing rules. By contrast, replacing those rules or the original task contract changes the context. A proposal with no established interpretation has a typing obligation before it belongs to D ; it is not automatically a completed anchor variation.

Theorem 4.2 (Exclusive effect exhaustion). *Every incidence in V has exactly one of the three effects. A variation within one original context that changes a required outcome has tensor effect. A skin variation preserves every required original outcome, including its semantic definedness. No fourth effect on those covered original outcomes remains.*

Proof. First compare the context indices. Inequality gives the first case and excludes the other two. If they agree, compare the complete profiles. Inequality gives the second case; equality gives the third. These conditions are disjoint and cover every pair. Each original totalized outcome is a coordinate of its profile, so a changed outcome implies unequal profiles. Conversely, equality of profiles gives equality of every original coordinate. An alleged additional effect with a specified original meaning must either change the context or some original outcome; the stated alternatives already locate that change. Coverage of those outcomes supplies the conclusion, rather than the availability of three names. $\square$

The three values classify *effects of variations*. They are not three standing values for one original qualification. Nor does a tensor-effect classification reduce the number of independent quantities within tensor content. A complete profile may have arbitrarily many coordinates.

Several content channels in one raw component. Let $D = \{0,1\}^2$ at one context, with required outputs $f_1(x,y) = x$ and $f_2(x,y) = y$. Treat the pair as one raw component. Its two coordinates can change independently, although both kinds of variation have tensor effect. Combining those outputs into one pair-valued profile gives a complete channel; it does not prove that its coordinates are functionally dependent. The theorem establishes one effect type per specified variation, not one scalar degree of freedom per raw component.

4.2 The canonical homogeneous refinement

Definition 4.3 (Homogeneous component class). For $c \in C$, put $V_c = \pi^{-1}(\{c\})$. A *homogeneous component class* is a nonempty subset $U \subseteq V_c$ on which ϵ is constant. Its effect is that constant value. Define the refined component carrier

$$C^* = \{(c, e) \in C \times \{A, T, S\} : \exists v \in V \pi(v) = c, \epsilon(v) = e\}$$

and $r : V \rightarrow C^*$ by $r(v) = (\pi(v), \epsilon(v))$.

Theorem 4.4 (Canonical refinement of arbitrary raw components). *The nonempty fibres of r are precisely the maximal homogeneous component classes. They partition V .*

Moreover, let $z : V \rightarrow Z$ be any classification that retains component identity and determines effect: there are functions

$$a : z(V) \rightarrow C, \quad b : z(V) \rightarrow \{A, T, S\}$$

with $\pi = a \circ z$ and $\epsilon = b \circ z$. There is a unique surjection

$$u : z(V) \rightarrow C^*, \quad r = u \circ z.$$

It is a bijection exactly when $r(v) = r(w)$ implies $z(v) = z(w)$. Thus r is the unique minimal homogeneous refinement, up to the unique commuting bijection over the original incidences.

Proof. For $(c, e) \in C^*$, the fibre $r^{-1}(\{(c, e)\})$ is nonempty, belongs to V_c , and has constant effect e . Every homogeneous subset of V_c with effect e is contained in that fibre. It is therefore maximal, and any maximal homogeneous subset must be that whole fibre. Fibres of a function partition its domain.

Define $u(z(v)) = (a(z(v)), b(z(v)))$. This is well defined on the attained image and equals $r(v)$. Every member of C^* is attained by r , so u is surjective. The equation $r = u \circ z$ forces its value at every point of $z(V)$, proving uniqueness. If the additional implication holds and $u(z(v)) = u(z(w))$, then $r(v) = r(w)$ and hence $z(v) = z(w)$, giving injectivity. Conversely, injectivity of u and $r = u \circ z$ give that implication. This also proves the stated uniqueness of a minimal refinement. $\square$

Corollary 4.5 (Exact condition for a pure raw component). *A raw component c has one realized exclusive effect precisely when $V_c \neq \emptyset$ and $\epsilon(V_c)$ is a singleton. In that case its entire variation family is one maximal homogeneous class. If $\epsilon(V_c)$ has several elements, its canonical refinement has exactly one nonempty class for each of those effects. If $V_c = \emptyset$, no realized effect is inferred.*

Proof. The asserted condition is exactly nonempty constancy of ϵ on V_c . The maximal classes supplied by Theorem 4.4 are indexed by its attained effects. The empty family has none. $\square$

This is a reconstruction of the typing problem at its actual incidences. It does not split a physical object or a syntactic field into independently recombinable pieces. A raw component can carry several kinds of work in different uses; the refined classes say exactly where each kind occurs. A claim that a particular component is pure must establish the corollary's universal condition from its actual formation and replacement rules.

A mixed component with all three effects. Take $D = \{0, 1\}^3$, let $\kappa(a, t, s) = a$ and $p_a(a, t, s) = t$, and use one raw component name. Include the variations

$$(0, 0, 0) \rightsquigarrow (1, 0, 0), \quad (0, 0, 0) \rightsquigarrow (0, 1, 0), \quad (0, 0, 0) \rightsquigarrow (0, 0, 1).$$

Their effects are respectively anchor, tensor, and skin. The raw component is mixed, while its three incidence classes are exclusively typed. These variations are all interpreted; the example does not depend on treating an ill-typed replacement as a completed change of context.

4.3 Exact minimality of the effect classifier

Definition 4.6 (Adequate effect classifier). Put $E = \epsilon(V)$. A classifier $z : V \rightarrow Z$ is adequate for effect typing when some $\delta : z(V) \rightarrow E$ satisfies $\epsilon = \delta \circ z$. It may retain more information than the effect. It is *minimal complete* when it is adequate and also identifies every pair of incidences having the same effect.

Theorem 4.7 (Unique minimal role classifier). *Every adequate effect classifier maps onto E through a unique decoder. Its attained image has at least $|E|$ elements. The quotient of V by equality of ϵ is the unique minimal complete effect classifier up to the unique commuting bijection.*

Since $|E| \leq 3$, an adequate classifier has the smallest possible number of attained states exactly when it has $|E|$ states; in that case its decoder is a bijection. Exactly three distinguishable states are necessary and sufficient if all three effects occur. Empty or smaller effect families have the corresponding smaller minimum.

Proof. Adequacy provides a decoder whose value at $z(v)$ is forced to be $\epsilon(v)$, so it is unique and onto E . Select one preimage of each member of the finite set E ; distinct effect values require distinct preimages. This proves the lower bound. The map $\epsilon : V \rightarrow E$ itself attains the bound.

If a classifier additionally identifies equal effects, its kernel equals that of ϵ . The map $z(v) \mapsto \epsilon(v)$ is then both injective and surjective, with inverse $\epsilon(v) \mapsto z(v)$. The equations force both maps on their attained images. More generally, an adequate classifier with $|E|$ attained states has a surjection between finite sets of equal size and hence a bijection. Conversely a bijective decoder gives equal kernels and exactly $|E|$ attained states. When V is empty both attained images are empty and the same statements hold. $\square$

An adequate classifier can split a tensor class into subclasses without creating another kind of original authority. For example, a classification may record both the tensor effect and which original coordinate changed. Forgetting that additional label returns the effect classifier. Its refinement is bookkeeping *relative to the task of identifying the effect type*; the underlying changed coordinate

remains genuine tensor content for the original object. There is no inference that every richer invariant classification is false. The exact uniqueness claim concerns minimal complete effect classifiers.

Proposition 4.8 (When all three effects are realized). *For the unrestricted nontrivial pair family on D , the following criteria hold:*

1. *an anchor effect occurs exactly when $\kappa(D)$ has at least two members;*
2. *a tensor effect occurs exactly when some Q_k has at least two members;*
3. *a skin effect occurs exactly when some realization fibre $R_{k,q}$ has at least two members.*

For a restricted incidence family V , each criterion must instead be witnessed by a permitted pair in that family. The size conditions alone need not provide such a pair.

Proof. Different attained contexts have witnesses in D , whose pair has anchor effect; any anchor pair conversely supplies two such contexts. Distinct attained profiles in one context have representatives in D_k , giving a tensor pair, and a tensor pair supplies those two profiles. Distinct points in one realization fibre give a nontrivial skin pair, while every skin pair lies in one such fibre. Restricting V can remove any of these witness pairs without changing the underlying carriers, which proves the final qualification. $\square$

For a problem with one fixed context, no change of governing context belongs to its internal variation repertoire. That fact does not make its preconditions optional; it means they are held fixed. Likewise a task with singleton realization fibres has no nontrivial skin variation, while every actual description still realizes its content. The exact minimality result tracks the work and effects that the specified target actually requires.

5 Functional deletion, realization, and independent factorization

Minimality must identify the task whose information or realizations are to be preserved. The minimal complete content interface, the minimal classifier of variation effects, and a minimal set of realization witnesses answer different questions. The preceding effect theorem does not establish a lower bound on the number of stored fields. This section gives exact tests for deletion and reconstruction, then characterizes when the dependent normal form admits genuinely independent Cartesian coordinates.

5.1 An exact test for deleting or merging information

Theorem 5.1 (Functional deletion alternative). *Let $F : D \rightarrow Y$ be an independently specified required outcome profile, totalized when definedness is relevant. Let $h : D \rightarrow H$ be the code retained after a proposed deletion, merger, or recoding. Exactly one of the following holds:*

1. *there is a unique decoder $\bar{F} : h(D) \rightarrow F(D)$ satisfying $F = \bar{F} \circ h$;*
2. *there are $d, d' \in D$ with $h(d) = h(d')$ and $F(d) \neq F(d')$.*

Thus the deletion preserves the required work precisely when $\ker h \subseteq \ker F$. In the second case, no procedure whose only input is the retained code can always recover the required original result.

Proof. A decoder sends equal retained inputs to equal required outputs, excluding the second alternative. Conversely, suppose there is no pair of the second kind. Define $\bar{F}(h(d)) = F(d)$. The absence of such a pair makes this well defined; the attained image supplies its codomain and forces its value everywhere, proving uniqueness. Negating the required constancy gives the displayed pair. At its shared retained input, a universally correct decoder would have to return both distinct required outputs, which is impossible for a function. $\square$

For an indexed family of required outcomes, take their complete profile as F . Unequal profiles then supply an original coordinate witnessing the loss. A partial result can fail this test solely through its definedness, even when every pair of defined values agrees. The totalized outcome retains that distinction.

Corollary 5.2 (Necessary work survives every faithful encoding). *Suppose a proposed reduced interface is claimed to preserve a specified original function. If it admits the second witness in Theorem 5.1, the claimed preservation fails. If it admits the decoder instead, the function remains represented by that decoder, including when its former field name has disappeared. Combining several required functions in one adequate code does not remove their work.*

Proof. The first assertion is the impossibility alternative. In the other case the equality $F = \bar{F} \circ h$ explicitly represents the original function using the retained code. A tuple of several functions is itself a profile, so the same reasoning applies to their combined encoding. $\square$

This gives a testable meaning to functional necessity. The required question and its interpretation must survive; a particular field layout need not. In the dependent reconstruction, the tagged content $P(d)$ already determines $\kappa(d)$, and the raw realization d determines $P(d)$. Removing an additional written copy of a determined index is not a deletion of its function. Conversely, an encoding that actually loses a required index or outcome has the collision witness above.

Proposition 5.3 (No unrestricted lower bound of three stored fields). *Under arbitrary set-valued encodings, every exact representation by three coordinates can be encoded in one coordinate without losing its reconstruction. Consequently the existence of three necessary kinds of work does not by itself imply that three independently stored fields are necessary.*

Proof. Let an exact code take values in a specified subset $J \subseteq A \times T \times S$, with its assembly inverse. Regard J itself as the value set of one coordinate and retain that same code. The assembly inverse and all coordinate projections remain available, so no represented function or reconstruction is lost. A lower bound on the number of fields must therefore impose and justify a more restricted encoding class. Theorem 4.7, by contrast, counts distinguishable effect states and is unchanged by this repackaging. $\square$

5.2 Semantic preservation and realization coverage

Theorem 5.4 (Forgetting surplus versus recovering the original description). *The complete semantic code $P : D \rightarrow P(D)$ preserves every original outcome in its context. It permits recovery of the original description by a map $s : P(D) \rightarrow D$ satisfying $s \circ P = \text{id}_D$ exactly when every realization fibre $R_{k,q}$ is a singleton.*

More generally, an arbitrary retained code $h : D \rightarrow H$ permits exact original-description recovery by a map $s : h(D) \rightarrow D$ exactly when h is injective. A nontrivial fibre gives an explicit obstruction.

Proof. Every original outcome is recovered from the appropriate coordinate of P , so semantic preservation holds. If $s \circ P = \text{id}_D$ and $P(d) = P(d')$, applying s gives $d = d'$. Thus every attained fibre is a singleton. Conversely, singleton fibres define the inverse s on $P(D)$ and give the displayed identity. Apply the same argument with h and its attained image for the general claim. Two distinct descriptions with equal retained code are the required obstruction. $\square$

The original description and the original object need not have the same equality relation. Two lawful descriptions can realize one object. The theorem concerns recovery of the description when that is the requested task; it does not turn an unobserved typographical distinction into a new physical identity. If the original task genuinely requires an additional identity distinction, that question belongs in its complete profile.

Theorem 5.5 (Exact realization coverage). *Let $C \subseteq D$ be a proposed retained family of actual realizations. It realizes every attained original content exactly when*

$$P(C) = P(D), \quad \text{equivalently} \quad C \cap R_{k,q} \neq \emptyset \quad \text{for every } k \in \kappa(D), \ q \in Q_k.$$

Failure has a content whose realization fibre has no member in C . For a prescribed subfamily $B_0 \subseteq P(D)$ of required contents, the corresponding condition is $B_0 \subseteq P(C)$.

Proof. A content (k, q) belongs to $P(C)$ precisely when some $c \in C$ has context k and profile q , which is precisely membership in $C \cap R_{k,q}$. This proves the equivalences and the witness on negation. Restricting the quantified contents gives the last assertion. $\square$

If two representatives of a fibre are interchangeable for every original use, retaining both is unnecessary for mere realization coverage. Removing the last available representative of a required content is different: its required realization is then missing. A mathematical description of the content and a supplied realization witness are distinct deliverables. The theorem identifies their relation without treating surplus choice as additional authority.

Corollary 5.6 (Exact minimality of a realization family). *A family $C \subseteq D$ that covers $P(D)$ is inclusion-minimal among covering families if and only if it contains exactly one member of each attained realization fibre. Such a family exists exactly when a section $s : P(D) \rightarrow D$ of P exists. An explicit section supplies such a family by its image.*

Proof. Coverage supplies at least one member of every fibre. If C contains two distinct members of one fibre, removing one leaves coverage unchanged, so C is not minimal. If it contains exactly one member of each fibre, removing any member loses that fibre and hence coverage. A section chooses one member of each fibre and is injective because $P \circ s = \text{id}_{P(D)}$. Conversely, a covering family with one member in each fibre defines the unique section taking each content to that member. $\square$

Existence of a section is not part of the choice-free dependent reconstruction theorem. It is an additional uniform selection task. An exhibited finite selection can supply it; a set-theoretic choice principle can establish abstract existence in a suitable foundation. Neither alone proves that the selector is computable, licensed for an intended operation, or determined naturally by the original content. A single requested realization needs a witness in its own fibre; it need not solve the selection problem for every content simultaneously.

One carrier, two distinct deletion tasks. Let $D = \{0, 1\}^3$ with $\kappa(a, t, s) = a$ and $p_a(a, t, s) = t$. The complete original profile is (a, t) , while raw-description recovery asks for (a, t, s) .

- Forgetting a and retaining (t, s) loses the original context: $(0, 0, 0)$ and $(1, 0, 0)$ collide.
- Forgetting t and retaining (a, s) loses original content: $(0, 0, 0)$ and $(0, 1, 0)$ collide.
- Forgetting s and retaining (a, t) preserves the complete original profile, but loses raw-description recovery: $(0, 0, 0)$ and $(0, 0, 1)$ collide.

The family $C = \{(a, t, 0) : a, t \in \{0, 1\}\}$ is an inclusion-minimal realization cover. It retains one lawful witness of each content while deleting representational alternatives. If the original task instead requires a result distinguishing s , that result must be included in the profile; its variation then carries content for that task. It cannot be both invisible to every required outcome and indispensable to preserving those same outcomes.

5.3 Reconstruction on a compatible domain

The canonical decomposition is dependent: the possible contents can depend on the context, and the realization fibre can depend on the content. To claim a separate product factorization, specify the proposed coordinates and their actual joint domain.

Theorem 5.7 (Exact coordinate reconstruction criterion). *Let A, T, S be proposed coordinate sets, let $J \subseteq A \times T \times S$ be their declared compatible tuples, and let $e : D \rightarrow A \times T \times S$ be the proposed decomposition map. There exists an assembly map $m : J \rightarrow D$ satisfying*

$$e(D) \subseteq J, \quad m(e(d)) = d \ (d \in D), \quad e(m(j)) = j \ (j \in J)$$

if and only if e is injective and $e(D) = J$. When it exists, m is unique.

Proof. The first inverse equation makes e injective. The second puts every $j \in J$ in $e(D)$, and the displayed containment gives the reverse inclusion. Conversely, injectivity and image equality define $m(j)$ as the unique d with $e(d) = j$. Both inverse equations follow, and the same characterization forces uniqueness. $\square$

The set J can be specified by independently given compatibility conditions and tested against $e(D)$. Defining $J = e(D)$ is also legitimate, but then equality of images is definitional and supplies no new compatibility theorem. Injectivity still requires proof. Neither a list of three coordinates nor a surjective assembly alone proves unique reconstruction.

Corollary 5.8 (Independent recombination is rectangularity). *Suppose A, T, S are the individually attained coordinate sets and Theorem 5.7 holds. Every combination of individually available coordinate values has exactly one realization if and only if $J = A \times T \times S$.*

Proof. Every such combination belongs to the product. Existence of its realization is equivalent to its membership in $e(D) = J$, and uniqueness follows from injectivity. Thus all combinations are realized precisely when the product is contained in J ; the reverse containment is given. $\square$

Compatibility can obstruct a product. Take $D = \{(0, 0), (1, 1)\}$ and $e(x, y) = (x, y, *)$. Both first and second coordinates attain both Boolean values, but the joint tuples $(0, 1, *)$ and $(1, 0, *)$ are absent. The code reconstructs perfectly on its diagonal domain, while independent recombination fails. Individually available factors therefore do not imply a rectangular joint domain.

A product image can still lose reconstruction. Take $D = \{d_0, d_1\}$ and let all three proposed coordinates be constant. Their product has one point, and every proposed tuple is realized. The code nevertheless identifies d_0 and d_1 , so no inverse reconstructs both. Rectangularity and injectivity are separate obligations.

5.4 When the dependent normal form becomes Cartesian

Theorem 5.9 (Exact criterion for an ATS-compatible product). *Put $K_0 = \kappa(D)$, and let T, S be nonempty sets. The following are equivalent:*

1. *There is a bijection $E : D \rightarrow K_0 \times T \times S$ whose first coordinate is κ and whose second coordinate, written E_T , satisfies*

$$E_T(d) = E_T(d') \iff p_k(d) = p_k(d') \quad (d, d' \in D_k).$$

2. *There are specified bijections*

$$\theta_k : Q_k \longrightarrow T, \quad \eta_{k,q} : R_{k,q} \longrightarrow S \quad (k \in K_0, q \in Q_k).$$

Given the latter maps, the product reconstruction is

$$E(d) = (k, \theta_k(p_k(d)), \eta_{k,p_k(d)}(d)), \quad k = \kappa(d).$$

It is uniquely determined by those supplied maps. Their existence or uniqueness does not follow merely from the dependent reconstruction.

Proof. Assume (i). Define $\theta_k(p_k(d)) = E_T(d)$. The stated equivalence makes this well defined and injective. For any $t \in T$, choose an element $s_0 \in S$; surjectivity of E gives a preimage of (k, t, s_0) , proving surjectivity of θ_k . Define $\eta_{k,q}(d)$ to be the third coordinate of $E(d)$ on $R_{k,q}$. If two such third coordinates agree, their first and second coordinates also agree, so injectivity of E gives equal descriptions. For any $s \in S$, the preimage of $(k, \theta_k(q), s)$ has context k and profile q , proving surjectivity of $\eta_{k,q}$.

Conversely, assume (ii) and use the displayed formula. To reconstruct (k, t, s) , first take $q = \theta_k^{-1}(t)$ and then $d = \eta_{k,q}^{-1}(s)$. Membership in the fibre gives $\kappa(d) = k$ and $p_k(d) = q$, so these formulas are two-sided inverses. The second-coordinate condition follows from injectivity of θ_k . The supplied bijections force each coordinate of E , establishing the relative uniqueness. If D is empty, K_0 is empty and both families and the product reconstruction are empty; the same equivalence holds. $\square$

The requirement uses supplied compatible identifications, not an unexplained claim that all fibres have the same size. The resulting coordinate changes can still require re-realization in another fibre. A licensed physical or syntactic operation implementing such a change needs its own applicability and preservation proof. A mathematical product bijection does not authorize that operation.

Unequal realization fibres. At one context, let $Q = \{0, 1\}$, with one realization of 0 and two realizations of 1. Then D has three members. There is no bijection of the compatible form $D \cong Q \times S$: if S has one member the product has two, while if it has at least two the product has at least four. Nevertheless the dependent reconstruction exists exactly as stated. It represents the actual fibre geometry instead of adding missing Cartesian combinations.

Even when a product exists, its coordinates need not be canonical. For $D = K_0 \times T \times \{0, 1\}$, globally exchanging the last two labels gives a second equally valid product presentation. The commuting comparison between two specified presentations can be unique while the choice of presentation is not.

5.5 What the minimality and independence results establish

Theorems 4.7 and 5.1 establish exact role-necessity criteria. Different realized effect types cannot be identified by an adequate effect classifier. A required original distinction cannot be deleted from a faithful realization unless it remains recoverable from what is retained. A promised realization cannot be supplied by a family missing its entire required fibre.

Those statements do not identify necessity with independent storage. Cartesian factorization requires the joint reconstruction and rectangularity tests, or the compatible family of bijections in Theorem 5.9. The canonical dependent form applies before those extra conditions are proved. Its generality permits shared information, mixed raw components, unequal realization fibres, and constrained assemblies while retaining the exact original work at every stage.

6 Skin transformations and joint construction

6.1 All transformations within realization fibres

Skin equivalence need not mean replacing one printed name with another. It is equality of the complete original profile under fixed anchors. A transformation can alter a derivation, construction method, data structure, or execution plan and still preserve that profile when the original uses do not distinguish those features. Whether the transformation is an available, warranted operation is a further original question. We first classify its mathematical preservation property.

Theorem 6.1 (Complete characterization of neutral realization maps). *A function $n : D \rightarrow D$ preserves context and complete content, $Pn = P$, if and only if it consists of a family of functions*

$$n_{k,q} : R_{k,q} \longrightarrow R_{k,q}.$$

Restriction and assembly are inverse correspondences. Composition corresponds to fibrewise composition. Such an n is bijective if and only if every $n_{k,q}$ is bijective; in that case its inverse also preserves P .

Proof. If $Pn = P$, then $P(d) = (k, q)$ implies $P(n(d)) = (k, q)$, so n restricts to the stated maps. Conversely define $n(d) = n_{P(d)}(d)$ from a family of fibre maps. Each d lies in precisely one fibre, making this a total function with $Pn = P$. Both operations recover the original maps. Restriction commutes with composition. If n is bijective, its restriction is injective; any $e \in R_{k,q}$ has an inverse image d , and $P(d) = P(n(d)) = P(e)$, making the restriction surjective. Conversely bijective restrictions assemble their inverses into a global inverse preserving P . $\square$

Thus neutral endomorphisms form the product of the transformation monoids of the realization fibres; invertible ones form the product of their permutation groups. This describes every such map, not just a selected family of syntactic rewrites. The description does not assert that every set-theoretic map is computable, physically executable, or licensed. Those assertions have their own original warrant conditions. It establishes that a proposed transformation contributes no new original-answer authority exactly when it stays within the same content fibres.

Proposition 6.2 (Normalization and representative selection). *A function $n : D \rightarrow D$ satisfying both $Pn = P$ and*

$$P(d) = P(e) \implies n(d) = n(e)$$

has the unique form $n = sP$, where $s : T \rightarrow D$ is a section of P . Conversely every section gives such a function. Every such n is idempotent. It is injective if and only if all realization fibres are singletons.

Proof. Constancy on fibres gives a unique s with $sP = n$. Then $PsP = Pn = P$, and surjectivity of P gives $Ps = \text{id}_T$. Conversely $Ps = \text{id}_T$ gives $P(sP) = P$, and sP is constant on fibres. Idempotence follows from $sPsP = sP$. If a fibre contains distinct descriptions, their common image under n defeats injectivity. If every fibre is a singleton, P is bijective and its section is its inverse, giving $n = \text{id}_D$. $\square$

A section supplies a representative; it does not recover which original representative occurred. Canonical reconstruction in theorem 3.1 needs no section. Nor does the existence of a complete quotient force an invariant section to exist. If a fibre has two or more members, a purported selection invariant under every permutation within that fibre is impossible: swap its selected member with another while fixing all other fibres. The content remains fixed and the selected member moves. A smaller symmetry family may admit a selection, but its fixed-point conditions then need proof, as in the AMetric selector theorem [2]. There is no blanket prohibition on calculational normal forms.

6.2 Joint compatibility is an original condition

Suppose descriptions of sorts $1, \dots, n$ have complete maps $P_i : D_i \rightarrow T_i$, and an output sort has $P_0 : D_0 \rightarrow T_0$. Fix an original partial constructor

$$C : J \longrightarrow D_0, \quad J \subseteq \prod_{i=1}^n D_i.$$

The joint domain J records actual compatibility, not the condition that every individual input separately qualifies. The product code is $H = \prod_i P_i$. We say J is *saturated* when $H(x) = H(y)$ implies $x \in J \Leftrightarrow y \in J$.

Theorem 6.3 (Exact ATS assembly descent). *There exists a partial constructor $\bar{C} : \prod_i T_i \rightarrow T_0$ whose totalized answers satisfy*

$$\bar{C}^\perp(H(x)) = (P_0 C)^\perp(x) \quad \text{for every } x \in \prod_i D_i$$

if and only if J is saturated and

$$x, y \in J, H(x) = H(y) \implies P_0(C(x)) = P_0(C(y)).$$

On the attained product image the descended constructor is unique. Its domain there is $H(J)$. For finite n , every tuple of attained coordinate values is attained, so this determines $\bar{C}$ on the full displayed product.

Proof. If a descended totalization exists, equal retained inputs give equal answers, including undefinedness. This yields both conditions. Conversely define $\bar{C}(H(x)) = P_0(C(x))$ for $x \in J$. Output agreement makes this well-defined. Saturation ensures that an outside-domain tuple with the same retained code cannot conflict with this definition; all other codes are undefined. Surjectivity onto the attained product image proves uniqueness. A finite tuple of attained coordinate values has a tuple of preimages by finite choice, which proves the last assertion. $\square$

The proof is the generic constructor criterion of Exhaustion, Theorem 4.8, displayed here to identify its precise ATS interface. Its additional role consequence is that valid skin replacements in several inputs can be composed without changing the output content *when these joint conditions hold*. They cannot be inferred from unary invariance alone.

For example, let $D_1 = D_2 = \{0, 1\}$ and let both retained maps be constant. Take $J = \{(0, 0), (1, 1)\}$. The inputs $(0, 0)$ and $(0, 1)$ have the same retained code; one is compatible and the other is not. This is a joint-domain collision. The repair is to retain the relevant relational distinction or to prove a different adequate interface. Calling both inputs individually admissible does not repair the lost equality condition.

Proposition 6.4 (Coverage by all single-coordinate contexts). *Let $G = (P_0C)^\perp$ on $\prod_{i=1}^n D_i$, with n finite. Then G is constant on H -fibres if and only if, for each i and every fixed tuple z of other coordinates, the unary contextual map $x_i \mapsto G(z[x_i])$ is constant on P_i -fibres.*

Proof. Forward implication holds because replacing one coordinate by an equal-code coordinate preserves H . Conversely let $H(x) = H(y)$. Replace the coordinates of x by those of y one at a time. Every intermediate tuple belongs to the full ambient product. The relevant contextual condition preserves G at each replacement, including its undefined marker. After n steps $G(x) = G(y)$. This is exactly totalized joint descent. $\square$

This proposition gives a constructive bridge from independently covered interaction tests to the joint criterion. It requires *all* fixed contexts relevant to the original constructor, or a proved generating basis for them. A test only against selected favourable partners cannot establish the universal condition. The theorem explains both why a full contextual profile suffices and why an arbitrary collection of local qualification bits can fail.

6.3 Finite assembly and its interpretation

Corollary 6.5 (Compositional preservation of ATS content). *Consider a many-sorted finite-term language with fixed constants and strict original constructors. Strictness means that an unavailable child makes the parent unavailable. Suppose every constructor satisfies theorem 6.3. Then evaluation of each finite term, including availability and retained output content, depends only on the retained leaf contents. If two equal retained leaf assignments yield different answers, some constructor instance has a joint-domain or retained-output collision.*

Proof. Induct on the term. A variable reads a retained leaf and a constant has its fixed retained value. At a constructor, the induction hypothesis gives equal availability and retained values for its

children. A missing child makes both evaluations unavailable. Otherwise theorem 6.3 gives equal constructor applicability and retained output. For a discrepancy, descend to a discrepant child if one exists. If no child differs, the current constructor itself violates one of the two conditions. A finite descent terminates. $\square$

This is the ATS application of Exhaustion, Theorem 4.9. It covers the declared formation rules. To cover an independently fixed original task, each original use must be interpreted by such a term followed by a retained-output reader, and that totalized composite must be proved equal to the original answer. A complete semantic profile is still available when no such finite language covers the task.

In particular, a nontrivial compatible family can exist without a full Cartesian product of components. Composition preserves content on the proved domain; it neither fills missing compatible tuples nor proves injectivity of assembly. The reconstruction and independence tests in Section 5 remain separate obligations. Distinct well-typed decompositions can yield the same composite, just as an identity transformation and two cancelling transformations can have equal net action.

7 Complete operator envelopes and finite-use identity

An operator envelope must retain the distinctions belonging to the operator identity being asserted. Two operations can have the same domain and preserve the same invariant while returning different values. Conversely, different constructions can supply the same mathematical operation. This section establishes the conditions under which a concrete envelope determines every original use, including composition.

A candidate specification, its licence, its applicability, and an occurrence of its execution are different subjects. A candidate can be examined even when its claim to a licence is false. An admitted operator meets the original licensing conditions. Its conditional input/output specification does not assert that an application occurred. Likewise an ordered sequence specifies a candidate composition; its existence does not assert execution. Every determinate specification, input, and occurrence already incurs the kernel roles. The following conditions concern faithful representation of their particular original uses.

7.1 The independently specified operator problem

For each original state sort s , let X_s be its carrier and let

$$q_s : X_s \rightarrow Q_s$$

be its retained original profile. Let $(\theta_{s,i} : Q_s \rightarrow V_{s,i})_{i \in I_s}$ be the separating family of original totalized readouts: different points of Q_s have different values of some $\theta_{s,i}$. These are the coordinates of the complete profile, restricted to its attained image. Scalar value sorts use the identity interface. Any required original state history, joint-use distinction, or resource condition belongs among these questions; a present qualification bit is not presumed to contain them.

Fix a typed family $\Omega_{s,t}$ of candidate specifications. Each $o \in \Omega_{s,t}$ has an independently specified conditional partial map

$$T_o : D_o \longrightarrow X_t, \quad D_o \subseteq X_s.$$

This map gives the specified result of application on its domain. The original problem also supplies target metadata $m_k(o)$, a licence or revocation outcome $\lambda_o(x)$, and required check outcomes $w_{o,j}(x)$. Their value sorts distinguish an ordinary negative result from semantic undefinedness. Whenever the identity or applicability of an obligation itself matters, its original membership or identity question is included in these data; equality of one currently observed verdict is insufficient.

The primitive original questions in this operator problem are the metadata, the domain predicate $d_o = 1_{D_o}$, the totalized licence and check outcomes, and the conditional result questions

$$r_{o,i}(x) = \begin{cases} \iota(\theta_{t,i}(q_t(T_o(x)))) & x \in D_o, \\ \perp_i & x \notin D_o. \end{cases}$$

Here $\perp_i$ is a new tagged absence outcome and ι is the value injection. A conditional specification question is a question about what the specified operation returns at that input; it is not evidence that the input was actually processed. If the original task concerns only authorized application behavior, its application domain is restricted accordingly and that restricted conditional behavior is the one retained. The original licence and applicability questions that the task separately requires are still retained.

Definition 7.1 (Primitive state adequacy and the basis envelope). For every $o : s \rightarrow t$, require:

1. D_o is saturated on q_s -fibres;
2. every original licence and check outcome is constant on those fibres, including its definedness;
3. if $q_s(x) = q_s(y)$ and $x, y \in D_o$, then

$$q_t(T_o(x)) = q_t(T_o(y)).$$

These conditions give descended data $\bar{D}_o = q_s(D_o)$, $\bar{\lambda}_o$, $\bar{w}_{o,j}$, and $\bar{T}_o : \bar{D}_o \rightarrow Q_t$. Define the basis envelope

$$E(o) = ((m_k(o))_k, \bar{D}_o, \bar{\lambda}_o, (\bar{w}_{o,j})_j, \bar{T}_o).$$

The indices and meanings of these fields are supplied by the original operator problem. A proposed envelope does not establish its own coverage by naming a field “complete.”

Lemma 7.2 (Unique primitive descent). *Under Definition 7.1, the descended fields exist uniquely on their attained domains, and*

$$q_s^{-1}(\bar{D}_o) = D_o, \quad \bar{T}_o \circ q_s|_{D_o} = q_t \circ T_o.$$

Each original conditional result question is recovered from the corresponding descended field. A failure of domain saturation, check constancy, or output agreement has a same-fibre pair witnessing the respective failure.

Proof. Put $\bar{T}_o(q_s(x)) = q_t(T_o(x))$ for $x \in D_o$. Output agreement makes this well defined; domain saturation gives the inverse-image equality. Define licence and check outcomes by their constant values on each fibre. The representative formulas and surjectivity force uniqueness. Applying $\theta_{t,i}$ on the descended domain and the distinct absence marker outside it recovers $r_{o,i}$. Negating one of the three conditions gives two inputs with the same retained code and different required domain, check, or retained-output behavior. $\square$

An original call rule may require, for example, applicability, a positive licence, and a stated finite family of successful checks. Its accepted domain is then

$$G_o = \{x \in D_o : \lambda_o(x) = 1 \text{ and all required checks hold}\}.$$

The rule and each check are independently fixed. A further joint or history-sensitive warrant must be included when required. If a task specifies an infinitary condition, that condition may be an original semantic question; a finite procedure for deciding it needs its own proof.

7.2 A finite language of original operator uses

The original use language has typed variables and original input constants, together with the following finite constructors:

1. metadata, domain, licence, check, and conditional-result inspection of the candidate operator;
2. a call of the candidate or a fixed declared operator on a typed input;
3. strict sequential binding of an obtained result to a continuation;
4. finite products and fixed typed postprocessing of already obtained values;
5. a conditional which evaluates its Boolean guard and then its selected branch;
6. a fixed joint constructor whose applicability and retained output descend through its retained input interfaces.

Every use has a finite expression tree. Calls and constructors use only their stated arguments and retained data. Extra tests of raw representative identity or hidden history are additional original questions, not implicit privileges of postprocessing.

Evaluation retains the original distinctions between unavailability, a failed check, and a successful result whenever the task makes those distinctions. Formally, a result sort is enlarged to a disjoint union of its successful values and its required failure tags; its retained interface applies q_s to successful states and preserves each tag. Strict binding propagates a failed prerequisite rather than fabricating a successful value. A failed guard is likewise propagated, and an ordinary Boolean guard selects exactly one branch. The propagation convention is fixed in the original interpretation.

Every fixed primitive operator satisfies the state conditions of Definition 7.1. Every joint constructor has a saturated original joint domain and agrees on retained outputs at equal retained input tuples. Its licence or other required original conditions are among its retained applicability data. These are actual premises of the composition result. Individual qualification of the inputs does not establish them.

Theorem 7.3 (Finite-use preservation and separating envelope identity). *For candidates $o, o' \in \Omega_{s,t}$ satisfying the stated primitive conditions, the following are equivalent:*

1. $E(o) = E(o')$;
2. *every use in the original finite language, on every common original input assignment, has equal required failure tags or equal successful results through its stated retained output interface after substituting o or o' for the candidate. Hence every original output readout agrees, including required domain, licence, and check results.*

Thus the envelope is a complete and separating invariant for identity under these original finite uses. This conclusion requires the primitive inspections listed above to be original uses. If some envelope coordinate is unobservable in the original problem, envelope equality can remain sufficient without being necessary.

Proof. Assume (i). A primitive inspection agrees by equality of the corresponding envelope fields and Lemma 7.2. A candidate call has the same original domain and check outcomes on equal retained input states, and equal retained results on success. A fixed operator has the corresponding property by its primitive adequacy conditions.

Induct on the finite use tree, allowing the two evaluations to receive state arguments with equal retained profiles and identical scalar arguments. Variables transmit those relations; fixed input constants agree. For a strict binding, the first expression either gives the same failure tag or gives equal retained successful results. In the latter case the induction hypothesis applies to the continuation with those results. Equal previously obtained values give equal fixed postprocessing results. Equal Boolean guards select the same branch, whose outcomes agree by induction; failure of a guard is propagated identically. For a joint constructor, the induction hypothesis supplies equal retained argument tuples. Domain saturation gives the same applicability and retained-output agreement gives equal results. The fixed failure rules give the same tags otherwise. This covers every constructor. Reading any original output coordinate from equal retained results yields equality of its original totalized answer, proving (ii).

Conversely, suppose the envelopes differ. Distinct metadata have a primitive metadata inspector as a witness. Distinct domains have an attained input with different domain answers. Distinct licence or check functions differ at an attained input and are separated by that primitive inspection. Finally, if $\bar{T}_o$ and $\bar{T}_{o'}$ differ despite equal domains, some attained input has different retained outputs. A separating original readout $\theta_{t,i}$ distinguishes those outputs, and the corresponding conditional-result inspection separates the candidates. Each witness is a finite original use. Thus (ii) fails whenever (i) fails. $\square$

The converse uses original questions, not an assumption that a physical observer can effectively execute every inspection. The theorem is semantic and its functions need not be computable. It also explains the exact boundary of an envelope claiming to represent a richer practice: every original use of that practice must have an interpretation-preserving expression in the language or be retained as a further original question. An unrepresented mechanism, timing, joint constraint, or historical use defeats that coverage claim.

Corollary 7.4 (Original-question coverage and stronger identity). *Let a_u be each independently required operator-use answer. If every a_u is proved equal to the totalized result of a finite use above, equality of envelopes preserves every a_u . If an independently specified operator identity map ν is separated by those original questions, equal envelopes imply equal ν . If, additionally, equal ν preserves every primitive inspector, then*

$$E(o) = E(o') \iff \nu(o) = \nu(o').$$

Proof. Apply Theorem 7.3 to each representing use and then its proved original interpretation. Separation gives equality of ν . Conversely, identity invariance of primitive metadata, domain, check, and conditional-output questions gives equality of their envelope fields; the output readouts separate retained outputs, so their equality determines $\bar{T}$ as well. $\square$

This exposes the two obligations for full operator identity: the envelope cannot omit a required distinction, and it cannot insert a distinction irrelevant to that identity. In particular, the result does not identify different actual mechanisms merely because their scalar outputs agree. Original identity questions must be retained when the claim concerns those mechanisms.

7.3 Substitution, sequence legality, and lawful rebracketing

For $o : s \rightarrow t$ and $p : t \rightarrow u$, the conditional composite has domain

$$D_{p \circ o} = \{x \in D_o : T_o(x) \in D_p\}$$

and returns $T_p(T_o(x))$. An original authorized sequence may additionally require a joint condition $J(o, p, x)$, giving

$$G_{p;o} = \{x \in G_o : T_o(x) \in G_p \text{ and } J(o, p, x)\}.$$

If forming an admitted composite specification requires its own licence, that licence is another original condition. It does not follow from possession of the two component licences alone.

Theorem 7.5 (Exact envelope substitution for sequential use). *Suppose $E(o) = E(o')$, $E(p) = E(p')$, the primitive state conditions hold, and the original joint condition has a proved factorization, for every relevant typed candidate pair (u, v) and every input x ,*

$$J(u, v, x) = \bar{J}(E(u), E(v), q_s(x)).$$

In particular, the same factorization holds for both (o, p) and (o', p') . Then the two sequential candidates have equal authorized domains and equal retained final results. Further sequence inspectors agree whenever their original interpretations are finite uses of the retained data and intermediate states. Under that coverage condition, the complete sequence envelopes agree.

Proof. The first envelope equality gives equal first applicability and warrant outcomes and equal retained first results. The second operator's domain saturation and check constancy give equal second applicability and warrants on those results. The displayed factorization makes the joint condition equal. Hence the sequential authorized domains agree. On that domain, the retained second result is determined by its descended map and the retained first result, so it agrees as well. Theorem 7.3 supplies equality of every further sequence inspector having the stated original interpretation. $\square$

Corollary 7.6 (Finite substitution and grouping invariance). *Under the corresponding conditions at every node, envelope-preserving replacements throughout a finite strict sequence preserve its complete original use. For one fixed ordered word, if original legality is determined by its stepwise domains and warrants together with grouping-independent original predicates on its prefixes, every parenthesization has the same totalized result and legality.*

Proof. Substitution follows by tree induction using Theorem 7.5. For grouping invariance, every parenthesization unfolds to the same conditional path $x_0, x_1, \dots, x_n$, with $x_i = T_{o_i}(x_{i-1})$ whenever available. Every grouping is defined precisely when all the same steps are defined. It checks the same original step and prefix predicates and returns the same retained endpoint. No assertion about changing the ordered word follows. $\square$

A grouping-sensitive resource, proof-object, or event-history question is not erased by the corollary: it would violate the grouping-independence hypothesis unless separately preserved. Structural sequence identity is also distinct from occurrence of execution.

7.4 Constructive skin and the limit of authority

Corollary 7.7 (Envelope-preserving construction changes). *A typed transformation R of candidate descriptions satisfying $E(R(o)) = E(o)$ preserves every original finite use above. Such transformations are closed under composition. They need not be invertible or consist of literal renaming. A faithful reversible transport additionally requires a proved inverse and the corresponding preservation identities.*

Proof. Theorem 7.3 gives preservation. For two such transformations, $E(R_2(R_1(o))) = E(R_1(o)) = E(o)$. Nothing in this equation implies injectivity; the next section gives a many-to-one construction. The inverse condition is additional data, not a consequence of equality of profiles. $\square$

A constraint can be used as data in a sound operator construction. For example, a functional relation can specify a map, and a matrix can represent a linear transformation. The typed distinction is that a statement of compatible values does not alone establish a licence, an application, or an execution event. The same mathematical carrier can occur in different uses whose obligations must be checked separately.

Proposition 7.8 (No unsupported operation or execution from constraint data). *Compatibility data alone do not in general determine a unique output map, establish its original licence, or entail an occurrence of execution. An independently justified construction may supply the missing map and licence; its conditional semantics still do not establish an execution occurrence.*

Proof. On $X = Y = \{0, 1\}$, the relation $C = X \times Y$ admits both constant maps, so compatibility does not determine a unique map. Keeping C and a selected map fixed while varying an independently specified licensing condition shows that the compatibility facts do not determine that condition. Finally, the same licensed specification is consistent with a record containing no execution and a record containing an execution. Their specification facts agree while the original occurrence predicate differs. If C is a functional total graph, that additional property does determine its mathematical map, but it still supplies neither the independent licensing condition nor an execution record. $\square$

These separating models do not make an actual domain's licence arbitrary. They show that an inference omitting its independently fixed conditions cannot establish it. An unsupported label cannot replace the required proof, while a sound constructor using constraint data can satisfy the actual operator obligations. Unique readout, unique operator identity, unique actual occurrence, and unique ATS component decomposition remain distinct conclusions.

8 Constructive realizations and separating examples

The examples show both a substantive skin-preserving construction and failures of incomplete operator descriptions. The distinction is always judged against the original questions. Equal present qualification alone does not establish equal uses, and different internal constructions need not represent different mathematical content.

8.1 A recursive construction and a closed expression

The original mathematical bearer is the triangular-number function on the natural numbers. Its questions are the value at each natural-number input, its totality and natural-number output type, and the declared finite arithmetic, comparison, and composition uses of those outputs. One presentation uses recursion:

$$S(0) = 0, \quad S(n+1) = S(n) + (n+1).$$

Another uses the expression

$$C(n) = \frac{n(n+1)}{2}.$$

The original arithmetic and recursion rules supply their mathematical warrants. A proof of the quotient's integrality is required for the second expression to have the asserted natural-number codomain.

Proposition 8.1 (A constructive skin change beyond renaming). *Both presentations define the same total map $\mathbb{N} \rightarrow \mathbb{N}$ and have equal complete envelopes for the stated function-bearing task. Every finite declared arithmetic or conditional use preserves this equality. Replacing the recursive presentation by the closed expression changes its construction while preserving its original function profile.*

Proof. Natural-number recursion defines S and gives its totality. Induction proves $2S(n) = n(n+1)$. The zero case is immediate. If the identity holds at n , then

$$\begin{aligned} 2S(n+1) &= 2S(n) + 2(n+1) \\ &= n(n+1) + 2(n+1) \\ &= (n+1)(n+2). \end{aligned}$$

Thus $n(n+1)$ is twice the natural number $S(n)$, so the displayed quotient is a natural number and $C(n) = S(n)$ at every input. Domains, target type, totality, and original output questions agree. Fixed arithmetic and comparison consumers preserve equal numbers; declared partial consumers must additionally retain their domain tests. The finite-use and sequence theorems then preserve the equality through every declared context satisfying those conditions. $\square$

For example, the original question whether the result at n exceeds a fixed threshold k has the same answer in both presentations. A subsequent operation permitted only on positive outputs has the same applicability in both: it is unavailable at $n = 0$ and available at every positive n . Its further original readout is equal when its own primitive descent conditions hold. This checks both a branch and a partial continuation, rather than only the first numerical answer.

The internal recursion has not been claimed to be one finite unrolling uniform in n . Its total function is established independently by induction. The finite-use theorem concerns external contexts in which that function is used. A question about source-code identity, runtime steps, intermediate states, energy, or a particular execution history can distinguish realizations and must be retained when it belongs to the original target. The proposition establishes equality of the presented mathematical function; it does not identify two physical program executions.

On the two-element presentation set $\{S, C\}$, the explicit normalization $N(S) = C$ and $N(C) = C$ preserves the complete function profile and is not injective. Hence substantive skin preservation can be many-to-one. It neither requires reversible recoding nor establishes an independent Cartesian skin factor.

8.2 Equal values can conceal a lost applicability condition

Let

$$T : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}, \quad T(x) = x/x, \quad U : \mathbb{R} \rightarrow \mathbb{R}, \quad U(x) = 1.$$

Their values agree wherever T is defined, but their original domain answers differ at zero. Their complete envelopes are therefore unequal. Restricting U to the same nonzero domain produces an equal conditional map for that restricted original task; asserting that the unrestricted map was already the same ignores an original question.

The example also prevents a false proof of skin neutrality by comparing only successful results. A failed or unavailable use is part of the original totalized profile when the task requires it.

8.3 Domain and invariant names do not determine an operator

On $\{0, 1\}$, the identity and Boolean negation are total and both preserve Boolean membership. They may share a positive licence and have no exceptional failure case. Nevertheless the original output query at 0 returns 0 for one and 1 for the other. A record containing only “Boolean domain, Boolean invariant, licensed” is not a complete operator envelope.

Conversely, suppose the conditional map is identity in both specifications but the original context-dependent licence is true for one and revoked for the other. Their numerical outputs agree, yet the original licensing question separates them. Neither direction can be repaired by treating a heading called “licence” or “invariant” as proof of its actual content.

8.4 A continuation distinguishes equally qualified states

Let states s, t, f have current qualification $b(s) = b(t) = 1$ and $b(f) = 0$. A declared repeat operation fixes s and sends t to f . The interface retaining only b identifies s and t , but their repeat results have different qualification. It fails the output agreement required for primitive descent.

If one claimed that two operator implementations were interchangeable because they both returned a currently qualified state, one could therefore be returning s and the other t . The original repeat-use question separates them. A complete output profile retains that distinction; equal current bits do not establish equal envelopes.

8.5 Joint licensing can defeat unary substitution

Two devices a, b implement the same Boolean map and satisfy the same unary applicability checks. Suppose the original sequence rule requires two different devices, so that

$$J(o, p) = 1 \iff o \neq p \quad (o, p \in \{a, b\}).$$

An envelope that forgets device identity gives equal retained tuples for (a, a) and (a, b) , while only the second pair satisfies J . The joint domain is not saturated. Theorem 7.5 correctly refuses substitution because its factorization condition for J fails.

Including the original device-identity question repairs this loss. It does not turn the physical identity into an optional interpretive overlay: the two devices were already distinct determinate objects. The repaired interface represents the distinction required by this original joint use.

8.6 Endpoint equality and retained sequence history

Boolean negation followed by negation has the identity endpoint map. If the original task asks only for that endpoint function and the independently justified domain and warrants, its extensional result agrees with identity. If the task also asks for the intermediate Boolean value or the number of calls in the retained sequence, it distinguishes the two constructions.

The sequence theorem retains such original inspectors when they belong to the target. Its grouping result concerns parenthesizations of one fixed ordered word with grouping-independent conditions. It does not claim that removing intermediate occurrences, changing order, or replacing a two-step construction by a one-step construction preserves every possible history question.

8.7 A nonseparating envelope and an unnecessarily fine envelope

A constant scalar readout on two different original operator identities has one value but does not establish a unique operator. Formally, let $\nu(o_0) \neq \nu(o_1)$ while $c(o_0) = c(o_1) = 0$. The readout fails the identity-separation condition of Corollary 7.4. To infer one original identity, the required identity questions must be determined too.

The opposite defect is an envelope augmented by an opaque tag that no original use can inspect. Two candidates with the same original answers but different tags then have different recorded envelopes. Equality of such envelopes may still suffice for use agreement, but it is no longer necessary. This is why the converse in Theorem 7.3 requires that its basis coordinates be original inspectable questions.

8.8 A licensed plan with no executed occurrence

A licensed identity specification and an admissible finite plan containing it can exist in a retained record with no execution occurrence. The same specification can also occur in a record of a completed use. Licence and sequence legality agree in the two cases; the original occurrence question does not. Execution is therefore neither inferred from licence nor inferred from writing the sequence.

This leaves room for a sound constructional warrant: the conditional result of an operator can be proved by its mathematical construction. A new actual occurrence, when claimed, still has its own bearer, conditions, and record. No original failed occurrence becomes successful merely because an equal-looking specification or a later successful occurrence is substituted for it.

9 Exhaustion, failure witnesses, and application criteria

9.1 Exact tests for ATS realizations

The theorem chain is closed by explicit alternatives. Each application identifies the object of its claim before applying a test. A claimed semantic interface must preserve the original context and answers. A role classifier must preserve the effects of the specified variation incidences. A realization must meet the required coverage, and a unique reconstruction must have two inverse maps on its declared domain.

Theorem 9.1 (Witnessed exhaustion of ATS realization claims). *Fix an original descriptive problem, a declared variation family, and a proposed representation with the claims listed below. Every false claim of one of these forms has the corresponding witness:*

1. Semantic adequacy: *equal retained codes with different original context or answer profile.*
2. Exclusive component typing: *an empty variation family when a realized role was claimed, or two permitted variations of one component with different effects.*
3. Effect-classifier adequacy: *equal classifier codes with different effects. If adequacy holds but minimality fails, two incidences of the same effect have different attained codes.*
4. Realization coverage: *a required content whose fibre contains no retained realization.*
5. Exact coordinate reconstruction: *an actual code outside the declared domain, a declared tuple with no original preimage, or two different descriptions with the same code.*
6. Independent recombination: *a tuple of individually attained coordinates absent from the actual compatible image.*
7. Joint semantic descent: *equal retained input tuples with different applicability or different retained output.*
8. Exact envelope identity for the declared finite language: *for a proposed envelope h , either equal h -codes with different original finite-use answers (failure of adequacy), or equal answers to every such use with different h -codes (failure of minimality). Under the hypotheses for the actual basis envelope E , neither defect occurs; a false asserted equality $E(o) = E(o')$ is separated by a differing primitive inspector. For an additional original identity map ν , either equal envelope codes with different ν -values or equal ν -values with different envelope codes witnesses failure of the claimed equivalence.*

The alternatives are exhaustive for these specified claims. They can overlap. Missing proofs of an interpretation, a semantic equality, or coverage are unestablished obligations; their absence alone is not a counterexample proving the claim false.

Proof. For (i), theorem 5.1 applied to P gives the exact alternative; different profiles differ in an original coordinate, unless the contexts already differ. For (ii), negate nonempty constancy in corollary 4.5. For (iii), adequacy is precisely constancy of the effect on classifier fibres; minimality additionally identifies equal effects, as proved in theorem 4.7. For (iv), use theorem 5.5. For (v), theorem 5.7 requires image equality and injectivity; either failed containment gives one of the first two witnesses, and failed injectivity gives the third. For (vi), use corollary 5.8. For (vii), use the two necessary and sufficient conditions in theorem 6.3. For (viii), negating equality of the kernels of h and the complete finite-use profile gives one of the two displayed pair types. The equivalence in theorem 7.3 rules out both for its actual basis envelope and supplies a primitive inspector whenever its fields differ. Negating either direction of the additional identity equivalence in corollary 7.4 gives the corresponding ν -pair. Each argument refers to the independently interpreted claim and not to a taxonomy imposed on its failures. $\square$

For the canonical constructions, the preceding proofs establish the conditions. For an offered replacement, the theorem localizes the mathematical challenge. A finer tensor classification that still decodes the original profile is an adequate enriched interface. It is a counterexample to an

unwarranted claim that all adequate interfaces are minimal, not a counterexample to the unique minimal interface theorem.

9.2 Failure localization is a dependency question

The role classification does not imply a universal finite number of admissibility levels. When a concrete argument has a finite dependency graph, its required premises and their truth values can be examined on that graph. The word “stress” can name a failed obligation, but adds neither a force law nor a unique-cause theorem.

Proposition 9.2 (Exact minimal-failure localization). *Let $(L, \prec)$ be a finite strict partial order of declared obligations, and let $B \subseteq L$ be its nonempty set of failed obligations. Then*

$$\text{Min}(B) = \{b \in B : \text{there is no } c \in B \text{ with } c \prec b\}$$

is nonempty, is an antichain, and every failed obligation has a member of $\text{Min}(B)$ at or below it. There is a unique minimal failed obligation exactly when this set is a singleton. A strict total order guarantees that singleton property, but does not identify a unique causal mechanism.

Proof. Starting at a failed obligation that is not minimal, choose a strictly smaller failed obligation. Finiteness and strictness force this descent to stop, yielding a minimal failed obligation below the starting one. Distinct minimal failed obligations cannot be comparable, since comparability would defeat minimality of the larger. The uniqueness criterion is the definition of a singleton. A finite strict total order has a least member in every nonempty subset. An order relation records the declared precedence only, so no claim about an independent notion of causation follows from these order facts. $\square$

For two independent failed premises feeding one conjunctive conclusion, both can be minimal. If a conclusion has alternative sufficient proof routes, failure of one route does not by itself defeat the other. Propagation requires the actual dependency rule: where validity requires every predecessor, a failed necessary predecessor invalidates that offered route. It is not a theorem that every reachable object in an arbitrary graph loses standing. This distinction keeps failure analysis tied to the proof that is actually being claimed.

9.3 Retention, new constructions, and the fixed incidence

Exhaustion, Theorem 6.2, distinguishes retention of an original negative occurrence from absorption of every later endpoint into negativity. A retained failed trial cannot be made a successful trial by renaming it. A new trial after repair has a new occurrence and can meet the same qualification. Likewise two negations can have an identity endpoint while a record of their intermediate states remains different from the record of a single identity operation. The complete profile includes whichever historical questions belong to the original task.

The same distinction governs a repair to a defective description. Adding the missing unit, proving a domain condition, or supplying a correct constructor can produce a new qualified description. It does not establish that the earlier incomplete description already met its obligations. Exhaustion prohibits evasion by unchanged-input postprocessing; it does not prohibit constructing the missing input, supplying a new proof, or carrying out a new admissible operation.

9.4 Dependent reconstruction and independent components

The elementary decomposition of a set into its fibres provides the basis for the canonical ATS construction. The theorem chain identifies the middle space through independently interpreted original uses, establishes its unique minimal universal property, reconstructs every alternative minimal presentation, refines mixed component incidences, and gives exact tests for deleting, recombining, and using the retained data. Independent components require the additional product conditions below.

A claimed concrete product must supply its coordinates and assembly. Theorem 5.9 states the exact extra structure needed to turn the dependent organization into that product. A cardinality match alone need not select a canonical or licensed identification. A specified family of fibre bijections does supply the mathematical reconstruction, and any claimed operational realization is then evaluated at its original domain and warrants.

9.5 Homogeneity of a whole component

A whole component has one realized role exactly when its actual declared variation family is nonempty and homogeneous. Corollary 4.5 is an exact theorem about that claim, and theorem 4.4 constructs the minimal homogeneous refinement in every case. A priority convention can assign one headline role to a mixed component, but the priority rule does not show that its other effects are absent. A tensor profile can contain many independent quantities without producing a fourth effect type. Both distinctions matter in downstream structural and physical applications.

9.6 Semantic preservation and realization requirements

Semantic preservation, exact description recovery, and realization coverage impose distinct requirements. Surplus distinctions between equivalent realizations can be discarded when only original semantic answers must be preserved. A claim to reconstruct the exact description must preserve its identity distinctions. A claim to supply a realization of each required content needs at least one member of every required fibre. Theorems 5.4 and 5.5 and corollary 5.6 prove these three facts together. They neither assign new authority to an invisible difference nor infer an actual supplied realization from a merely named content.

If a new physical question distinguishes two realizations, that difference is tensor content for the enlarged original question family. This does not change the necessity of the kernel governing each determinate realization. It changes what the representation is obliged to preserve.

9.7 Selectors, normal forms, and operational warrants

A selector may be an explicitly licensed construction, a mathematical choice, or a convenient normal form. Its availability, invariance, and operational use are distinct claims. A within-fibre normalization preserves content by proposition 6.2; it does not identify which representative previously occurred. A selector that changes a required future answer is a content-changing operation, and its warrant must establish that operation. A purported pre-fixational source of a selected determinate item is subject to the AMetric non-transmission and symmetry constraints. Nothing in the interior calculation supplies such boundary authority.

9.8 Semantic completeness and effective procedures

An effective decision procedure requires computational hypotheses in addition to set-indexed profiles, fibre tests, or a classical proof that a witness exists. A finite presentation can give a decidable test in a particular model, while a complete original question family can contain undecidable or infinitary questions. Finite program induction proves coverage of its given syntax. An eventual-success property can have no finite bad-prefix witness [8]. Such a property remains an original semantic coordinate if required; the finite calculus must not be claimed to decide it without an additional proof.

10 The reference theorem and downstream use

Theorem 10.1 (Canonical ATS decomposition and enforcement). *For every independently interpreted descriptive problem of definition 2.1, the following statements hold.*

1. *The original carrier has the canonical, choice-free dependent reconstruction*

$$D \cong \coprod_{k \in K} \coprod_{q \in Q_k} R_{k,q}, \quad D \twoheadrightarrow T \twoheadrightarrow K.$$

The content space is the unique minimal complete middle space up to the unique commuting bijection; faithful minimal alternative reconstructions have the same dependent fibre structure.

2. *Every specified nontrivial variation has exactly one anchor, tensor, or skin effect. Every component-incidence family has the canonical coarsest refinement into nonempty homogeneous role classes. Its minimal effect classifier has exactly as many attained states as there are realized effect types, and precisely three when all three occur.*
3. *Every required original function survives a proposed deletion exactly when it factors through the retained code. A failed deletion has an original distinction as its witness. Exact reconstruction and independent recombination are governed by injectivity, compatible-image equality, and rectangularity; an ATS-compatible Cartesian product exists exactly with the fibre identifications of theorem 5.9.*
4. *All neutral realization maps are exactly the fibrewise maps over T . Their invertible instances are exactly the fibrewise bijections. A supplied normalization can preserve content without preserving the identity of the original representative.*
5. *Joint and finite strict constructions preserve retained ATS content under their proved domain, output, and original-interpretation conditions. Unary adequacy does not supply a missing joint condition.*
6. *For an original operator problem with the inspectable basis and state conditions of Section 7, basis-envelope equality is equivalent to equality under every declared finite use. Its extension to additional original identities or uses requires the explicit coverage and separation conditions of corollary 7.4.*
7. *Every false realization claim of the forms in theorem 9.1 has its corresponding semantic, classification, compatibility, or reconstruction witness. Faithful recoding cannot erase that original mismatch; an actual correction must discharge the failed condition.*

The descriptions, states, records, carriers, and endpoints to which these constructions refer are necessarily kernel-governed as non-degenerate determinate objects. The representation tests specify how their original uses are preserved; they do not create an optional condition of kernel applicability.

Proof. Part (i) is theorems 3.1 and 3.3 and corollary 3.4. Part (ii) is theorems 4.2, 4.4 and 4.7. Part (iii) is theorems 5.1, 5.7 and 5.9 and corollary 5.8. Part (iv) is theorem 6.1 and proposition 6.2. Part (v) is theorem 6.3, proposition 6.4, and corollary 6.5, with original-use coverage as stated there. Part (vi) is theorem 7.3 and corollary 7.4. Part (vii) is theorem 9.1; preservation of the witness under faithful interpretations follows by proposition 2.2 and the corresponding commuting outcome maps. The final necessity statement imports the Kernel result [1]; none of the mathematical equivalences uses an assumption that the original determinate bearer can optionally evade that result. $\square$

10.1 What may be invoked

The table states the result together with what an application must establish. The interpretations are part of the proof, not an after-the-fact verbal identification.

Result	Required application proof	What it does not supply
Canonical reconstruction: theorem 3.1	The original context map and all claimed answer meanings.	Three freely recombinable or independently stored factors.
Unique content: theorem 3.3	Adequacy of the proposed middle map; minimality for a bijection.	Equality of enriched raw descriptions or a unique occupant.
Alternative realization: corollary 3.4	Minimal complete content and a faithful bijective assembly on the stated fibres.	A uniquely preferred coordinate system or a selected physical realization.
Pure roles: theorem 4.4 and corollary 4.5	The actual permitted variation family; nonempty homogeneity for a raw component.	Homogeneity of every symbol, clause, or composite object.
Three-role minimality: theorem 4.7	All three effect types realized if a three-state lower bound is claimed.	A lower bound of three written fields or independent degrees of freedom.
Functional necessity: theorem 5.1	The original required function and the code retained after deletion.	Loss of a function merely because its former name disappears.

Result	Required application proof	What it does not supply
Realization coverage: theorem 5.5	A supplied witness in every required fibre.	A uniform selector, effective construction, or natural preferred representative.
Independent factors: theorems 5.7 and 5.9	Injectivity, compatible-image equality, and the relevant coordinate or fibre identifications.	Compatibility of arbitrary combinations from unary facts.
Neutral construction: theorem 6.1	$Pn = P$ for the complete original profile; operational warrants if action is claimed.	Invertibility, execution, or authority over a question the profile omits.
Joint and finite use: theorem 6.3 and corollary 6.5	Joint-domain saturation, output agreement, and interpretation-preserving coverage of original uses.	Coverage of an arbitrary practice from a selected finite syntax.
Operator identity: theorem 7.3 and corollary 7.4	Inspectable primitive basis, state adequacy, original-use coverage, and identity separation where required.	Equal mechanisms, actual executions, or unique component decompositions from equal scalar readouts.
Failure localization: proposition 9.2	The actual dependency order and failed original obligations.	A universal level count or a unique causal source from a partial order.

10.2 Mathematical and physical applications

For a mathematical construction, the original task may fix a type, an extensional function, or an identity of proof objects. The same formula can have different realization obligations for those tasks. The triangular-number example proves equality of functions by induction and then transports that result through its declared uses. If proof identity is requested, the profile must preserve the distinctions defining that identity.

For a physical domain, determinate objects and records already incur the kernel. A concrete ATS application must still identify its original state questions, admissible transformations, units, occurrence identities, and joint conditions. Equations or experimental laws supply the domain facts; the present theorem organizes what a faithful realization must preserve. If the domain proves a unique numerical readout or a unique occupant, a complete interface conserves that proved conclusion. The existence of one scalar slot does not itself establish uniqueness across candidate realizations.

The anchor layer records which problem is being answered; the tensor layer contains every original distinction its uses require; and the realization layer records what actually presents that content. A purported new role that changes an original answer changes content or context. A refinement with no such effect remains realization information for those uses. Its independent interest as syntax, mechanism, or history can itself become the subject of another original question without generating a fourth effect type in the already fixed comparison.

10.3 Proof dependencies and verification

The quotient and generated-use results are supplied by Exhaustion; the necessity and pre-fixational results by the Kernel and AMetric papers. The role, reconstruction, minimality, and operator-identity results are proved here. Component independence, binary reduction, and occupant uniqueness each require separate domain proofs.

The accompanying project includes independent review records and finite model checks of instances and separating examples. The proofs establish the general results; the checks provide supplementary verification.

10.4 Conclusion

ATS gives a canonical organization of admissible description through explicit reconstruction maps. Its unique content layer preserves the full original task, its realization fibres retain what reconstruction requires, and its variation classifier separates context change, content change, and content-preserving realization change. The canonical refinement supplies exclusive roles wherever a coarse descriptive component mixes these effects. Deletion, recombination, and operator identity are governed by exact criteria with concrete witnesses on failure.

The decomposition theorem therefore supplies a mathematical witness for every loss of required original work in the stated task. No new authority is obtained from unchanged complete original content. The reconstruction criteria establish the precise conditions for independence and uniqueness.

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